

Strain Gauges

INDEX

LIST OF SYMBOLS.....	2
ACKNOWLEDGEMENTS	3
INTRODUCTION.....	3
1. HISTORICAL OVERVIEW.....	4
2.GENERAL.....	5
3. THE WHEATSTONE BRIDGE.....	7
4. SELF TEMPERATURE COMPENSATION.....	10
5. SIMPLE STRAIN GAUGE CONNECTIONS.....	12
6. THE BONDING OF STRAIN GAUGES.....	13
7. THE K-FACTOR.....	16
8. SELF-HEATING EFFECT OF STRAIN GAUGES:.....	17
9. STRAIN GAUGE BRIDGE AMPLIFIERS:.....	18
10. TWO WIRE VERSUS THREE - WIRE WIRING.....	21
11. SIMPLE SCALE.....	23
12. ROSETTE ANALYSIS.....	25
13. RESIDUAL STRESS ANALYSIS.....	27
14. TORQUE MEASUREMENT WITH FOUR STRAIN GAUGES.....	28
REFERENCES:.....	30
USEFUL BOOKS WHICH DEAL WITH STRAIN GAUGES AND THEIR USES:.....	30

LIST OF SYMBOLS.

b	- width (m)
d	- depth (m)
D	- diameter (m)
E	- modulus of elasticity (Young's Modulus) (N/m^2)
I	- electrical current (A)
K	- gauge- or K-factor of strain Gauges (dimensionless)
L	- length (m)
ΔL	- small change in length (m)
R	- resistance (ohm, Ω)
ΔR	- small change in resistance (ohm, Ω)
T	- torsion ($\text{N}\cdot\text{m}$)
V	- voltage (V)
W	- force (N)
x	- distance, length (m) (dimensionless)
β	- angle in degrees
ε	- strain (m/m or $10^6 \mu\text{m/m}$)
σ	- stress (N/m^2)
ν	- Poisson's ratio (dimensionless)
ρ	- resistivity ($\Omega\cdot\text{m}$)

ACKNOWLEDGEMENTS

These notes were originally compiled by Dr Terry (Eugene) Terblanche.

INTRODUCTION

This document is by no means comprehensive: to cover the entire field of strain gauges would take several volumes. In writing this we are attempting not only to cover the information that you will be acquiring in the lectures during the course, but also to instil an interest in the related field of experimental stress analysis.

We hope that by the end of the course you will be able to use strain gauges with confidence to solve the problems of strain analysis that you may experience in practice.

1. HISTORICAL OVERVIEW

As early as 1856 Prof William Thompson [1] (Lord Kelvin) had made known the results of experiments that he had conducted on wires that had been subjected to a particular strain. His finding was that the electrical resistance of the wires increases linearly with the increase in tensile stress in the wires.

More recently, in 1923, Bridgman [2] verified Prof Thompson's results. It was, however, not until late in the thirties that Simmons and Ruge [2], working independently with fine wires that were applied directly to the part being tested, obtained practical results for the first time from the experiments referred to above. This type of gauge was relatively unwieldy and had to be individually calibrated.

After the pioneering work described above the technology of strain gauges developed, rapidly with the so-called SR-4 strain gauge ("S" for Simmons and "R" for Ruge), the first really practicable strain gauge.

The strain gauge basically consists of very fine resistance wire which is laid out in a flat grid configuration (figure 1) or a helix configuration (figure 2).

With the development of photo-etching techniques foil strain gauges (figure 3) appeared on the market. Today roughly 99% of all strain gauges available are of this type.

A further development was the introduction of so-called semi-conductor strain gauges.

We shall not, however, deal with this type of strain gauge in this course.

2 . GENERAL

A strain gauge is basically a fine wire or metal foil which runs back and forth in order to fit a reasonable length of the conductor into a relatively short distance.

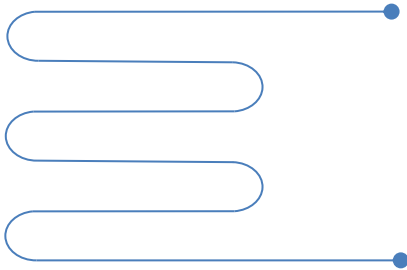


Figure 1. Metallic wire strain gauge

For examples, see figures 1, 2 or 3.

If this wire is brought into close mechanical contact with a metal, but is electrically isolated from it, the wire will deform with the metal and will be subjected to the same strain.

If the metal should be subjected to a tensile stress, the strain gauge would also be subjected to a tensile stress.

At this stage we can perhaps introduce the analogy of a tube. If a soft thin rubber tube is attached to a vessel with water in it, and the water is allowed to flow through the tube, a certain speed of flow will be maintained as long as the water pressure remains constant. If this tube is stretched out to double its original length, the speed of flow will decrease for two reasons. First the diameter of the opening decreases with the elongation, and secondly resistance to the flow through the tube will increase with the increase in length.

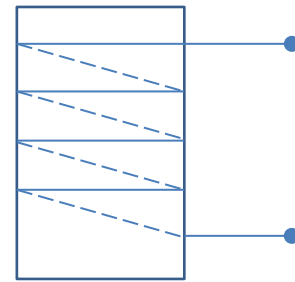


Figure 2. Coiled metallic wire strain gauge



Figure 3. Metal foil strain gauge

An electrical conductor that is elongated may be seen in a similar way. If the equation for the resistance, R , of a round conductor with length L , diameter D and resistivity ρ is given as

$$R = \frac{\rho L}{\frac{\pi}{4} D^2} \quad (1)$$

it can be seen that as the length, L , increases so will the resistance, R , become greater. Similarly, if the cross-section, dependent on D^2 , decreases, the resistance, R , will also become greater.

It can be shown from the above equation that

$$\frac{\Delta R}{R} = K \frac{\Delta L}{L} = K \epsilon \quad (2)$$

From equation (2) it is concluded that for a small change in length, ΔL , there is a small change in resistance ΔR (K , L and R being constant for one particular gauge).

Measurement by means of strain gauges is based, therefore, upon the determination of a specific change in length from a measured change in resistance, whether direct or indirect. It is clear from the strain gauge equation, equation (2), that there is a linear relationship between the change in resistance and the strain, $\epsilon = \frac{\Delta L}{L}$.

A strain gauge measures the average strain over its area, so to measure the strain at a point a strain gauge of zero length would be required. A strain gauge always has to have a finite length however, and theoretically a strain at a point can never be measured. In practice there are strain gauges on the market with an effective measuring length (the so-called grid length) of as little as 0,2 mm. For effective strain measurement with an error of less than $\pm 3\%$, strain gauge length is typically > 3 mm.

Naturally these short gauges are not always desirable because of anisotropy and heterogeneity of the material onto which they may be applied. An example is the case in which the strain in a concrete pillar must be determined. The stone aggregate which makes up a part of the concrete is subjected to a particular strain, whilst the matrix of sand and cement which surrounds the stones is subjected to another strain. If one wishes to obtain a good mean of the strain in this case, one should make use of a very long strain gauge which can run past several stones in the cement so that a mean may be obtained.

This example merely illustrates the fact that a good deal of attention should be paid to the application before deciding on a strain gauge. This is only one of the factors which are involved and which must be taken into account.

3. THE WHEATSTONE BRIDGE

Let us look at the basic strain gauge equation again.

$$\frac{\Delta R}{R} = K\varepsilon$$

Take the example of a $120\ \Omega$ strain gauge which is strained by approximately $1200\ \mu\text{m}/\text{m}$. (This is approximately the strain at yield point for mild steel.)

If the strain gauge factor is $K = 2$ then,

$$\begin{aligned}\Delta R &= RK\varepsilon \\ &= 120 \times 2 \times 1200 \times 10^{-6} \\ &= 0.288\ \Omega\end{aligned}$$

From this it is clear that if a strain gauge is applied to a part manufactured from mild steel, it would undergo only a $0.288\ \Omega$ change in resistance from no strain until the yield point. Assuming that the strain gauge was subjected to a tensile stress, then the resistance would increase from $120\ \Omega$ to $120.288\ \Omega$.

If it is necessary to accurately measure to $1\ \mu\text{m}/\text{m}$, then it must be possible to measure accurately to:

$$\frac{0.288}{1200} = 0.00024\ \Omega.$$

Most ohm-meters cannot come near this accuracy and as a result another method must be found.

As early as 1843, Sir Charles Wheatstone, the English scientist, developed the well-known Wheatstone bridge to determine a fourth unknown resistance with the aid of three known resistances. The Wheatstone bridge also lends itself to the measurement of very small changes in resistance. The principle on which this bridge works will only be briefly described. Study the accompanying sketch (Figure 4).

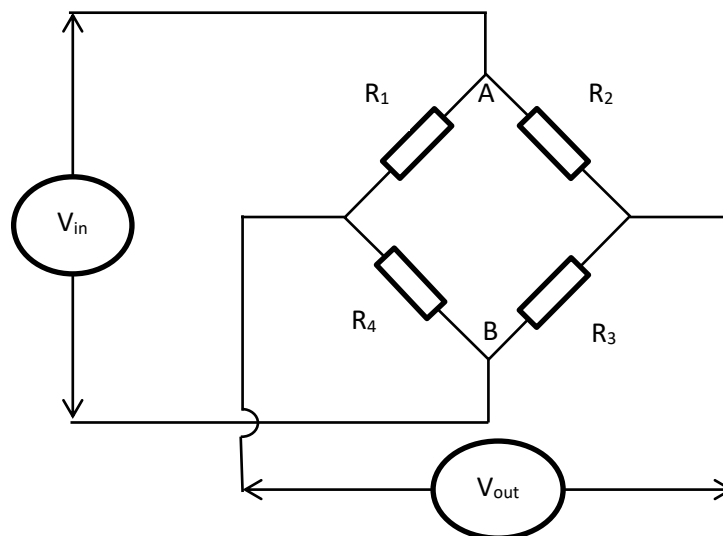


Figure 4. Wheatstone bridge configuration

If an input voltage V_{in} is applied between points A and B, an output voltage of V_{out} will be measured between points C and D which is dependent on the relationships between the resistances $R_1:R_4$ and $R_2:R_3$.

It is generally simpler to work with the relationship of the output voltage to the input voltage.

$$\frac{V_{out}}{V_{in}} = \frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_3} = \frac{R_1 R_3 - R_2 R_4}{(R_1 + R_4)(R_2 + R_3)} \quad (3)$$

If the bridge balances then

$$\frac{V_{out}}{V_{in}} = 0 \quad \text{for} \quad R_1 = R_2 = R_3 = R_4$$

or

$$R_1:R_4 = R_2:R_3$$

Generally speaking the bridge is used in this way:

First the zero-adjustment is made which means that the output voltage is zero. If the strain gauges now experience strain, and the change that every strain gauge experience is shown as ΔR_1 , ΔR_2 , etc., the output voltage will be given by the following equation. (Quadratic terms of ΔR that appear in the derivation are omitted as, for all practical purposes, they have little or no influence on the final result).

$$\frac{V_{out}}{V_{in}} = \frac{1}{4} \left[\frac{\Delta R_1}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4} \right] \quad (4)$$

If the strain gauge equation $\frac{\Delta R}{R} = K\varepsilon$ is substituted in the above, the following equation is obtained:

$$\frac{V_{out}}{V_{in}} = \frac{K}{4} [\varepsilon_1 - \varepsilon_2 + \varepsilon_3 - \varepsilon_4] \quad (5)$$

Many derivations can thus be made from this basic equation.

If for example only one strain gauge is active, and the others are all dummies, the following would hold true: (Dummies do not take part in the deformation that appears but are used purely to complete the Wheatstone Bridge).

$$\frac{V_{out}}{V_{in}} = \frac{K\varepsilon_1}{4}$$

Certain conclusions can be drawn from the basic equation. The terms ε_1 and ε_3 have the same sign, and ε_2 and ε_4 have the same sign. If all the deformations are the same magnitude, viz. $|\varepsilon_1| = |\varepsilon_2| = |\varepsilon_3| = |\varepsilon_4|$, no output voltage will then be obtained. The strain gauges should be set so that for instance R_1 and R_3 experience the same sign of strain, and R_2 and R_4 the same sign, but the opposite in sign from R_1 and R_3 .

In the case of a small cantilever beam with for example R_1 and R_3 applied above and R_2 and R_4 underneath as shown in the accompanying sketch, the following situation would hold as soon as the beam is strained by a point load W applied at the end.

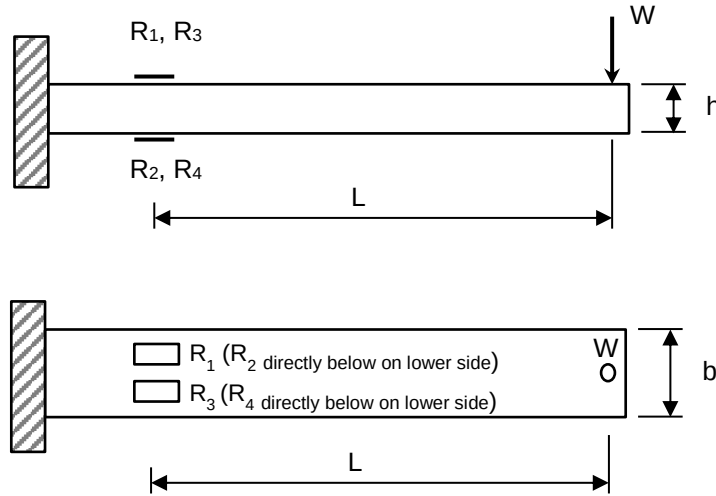


Figure 5. Cantilever beam with strain gauges full bridge configuration to measure strain due to bending

R_1 and R_3 are in an area of tensile strain which, by convention, are given a positive sign and R_2 and R_4 in an area of compression that has a negative sign.

If R_1 , R_3 , R_2 and R_4 are all bonded at the same distance from the force of W and the bar has a constant cross section, then ϵ_1 and ϵ_3 will be positive, and ϵ_2 and ϵ_4 negative. The value of the strain due to bending will be the same at each position, only the signs will differ, so

$$|\epsilon_1| = |\epsilon_2| = |\epsilon_3| = |\epsilon_4|.$$

Thus, if we define the strain due to bending as ϵ , the basic equation (equation 5) can then be written as follows:

$$\begin{aligned} \frac{V_{out}}{V_{in}} &= \frac{K}{4} [\epsilon - (-\epsilon) + \epsilon - (-\epsilon)] \\ &= \frac{K}{4} [4\epsilon] = K\epsilon. \end{aligned}$$

Thus the basic equation can be used to investigate several types of strain situations.

4. SELF TEMPERATURE COMPENSATION

The strain gauges that are in general use are usually manufactured from Constantan. (The approximate composition is 45% Nickel and 55% Copper.)

Supposing a strain gauge is applied to a steel part. If the steel exposed to fluctuations in temperature the steel will naturally expand with an increase in temperature and contract with a decrease in temperature. If the strain gauge does not stretch and shrink to the same degree it will naturally be subjected to a stress that will lead to a change in the resistance of the strain gauge.

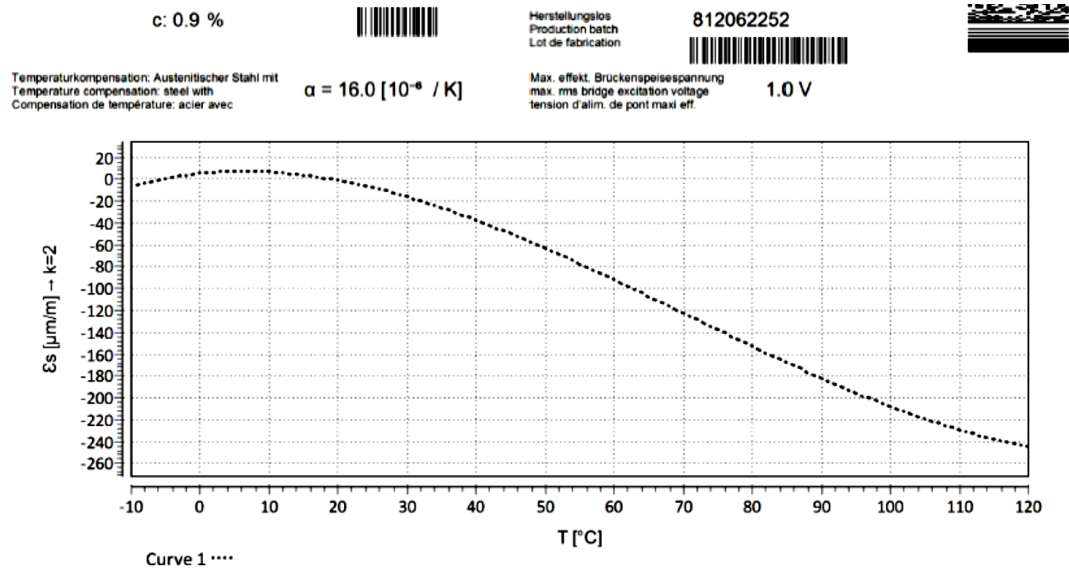
Strain gauges are thus manufactured with a coefficient of expansion that is the same as the metal to which it is to be bonded. This is done by adding small percentages of other elements to the basic composition given above.

MATERIAL	APPROXIMATE COEFFICIENT OF THERMAL EXPANSION $\mu\text{m}/\text{m}/^{\circ}\text{C}$
Steel and Martensitic Stainless Steel	11
Stainless Steel (Austenitic)	16
Aluminium	23
Magnesium	28

Strain gauges with other coefficients of expansion can be obtained from some manufactures by special request.

Unfortunately the coefficient of expansion of the strain gauge and the metal does not remain the same over a wide range of temperature. In addition other factors come into play, for example the so-called resistivity of the strain gauge, which does not increase linearly with temperature.

Figure 6 shows the apparent strain is plotted against the temperature of a strain gauge that is bonded to a material with the same coefficient of expansion, illustrates the above. The strain gauges are deliberately manufactured so that the horizontal section of the curve, i.e. where the apparent stain is zero or minimal, falls at approximately room temperature.



$$\epsilon_s(T) = 5.30 + 0.64 \cdot T - 5.28E-02 \cdot T^2 + 2.51E-04 \cdot T^3 \pm (T-20) \cdot 0.30 [\mu\text{m/m}]$$

Figure 6. Temperature compensation curve for strain gauge calibrated for steel. [excerpt from HBM datasheet]

5. SIMPLE STRAIN GAUGE CONNECTIONS.

A few of the simplest strain gauge connections are now briefly dealt with. Among others, there will be connections in which 1, 2 or 4 strain gauges are active. Assume that a bridge can have a maximum output of 4 if all 4 strain gauges are active.

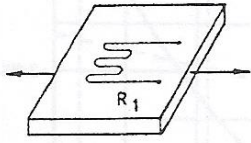
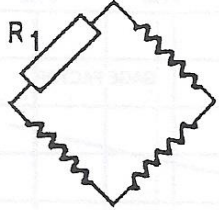
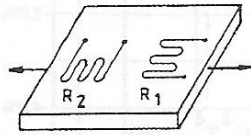
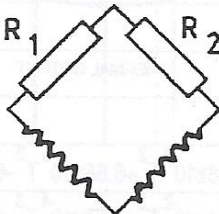
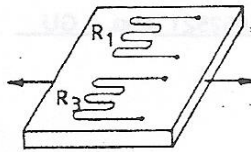
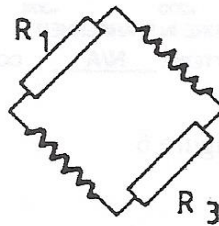
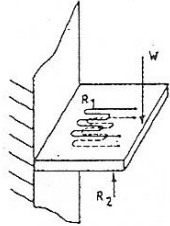
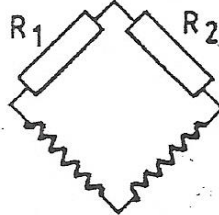
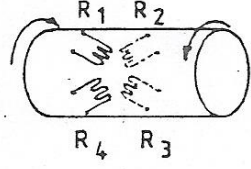
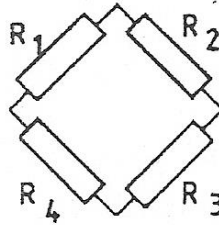
Application	Circuit strain gauge active shown in block	Output max. = 4	Remarks
		1	Not in the circuit, temperature compensated
		$1 + \nu$	Temperature compensated. So-called Poisson-connection
		2	Not Temperature Compensated
		2	Temperature Compensated. Tensile strain cancelled
		4	Temperature Compensated. Bending strain cancelled. Tensile strain cancelled

Figure 7 Simple strain gauge configurations

6. THE BONDING OF STRAIN GAUGES.

Today there are almost as many methods of bonding strain gauges as there are manufacturers of strain gauges, However there are certain basic steps that must be followed by all. The method that will be described here is also quoted in [3]. The type of glue, quickset (cyano-acrylite) or baked on glue(usually an epoxy-type glue)influences the gluing procedure. Our own practical experience over a number of years has proved the following to be a good method.

6.1 Preparation of the Surface:

A great deal of attention must be paid to this aspect, because it is unquestionably the most important step in the process of bonding strain gauges. Assume that the surface is fairly dirty and possible full of rust.

- (i) Remove rust and dirt with coarse (120 mesh) emery paper.
- (ii) Degrease the surface thoroughly with a degreaser, (Chlorothene NU, Freon, TF or even Carbontetrachloride). The cleaning agent must itself be very clean with very little residue.
- (iii) Use emery paper again (220-320 mesh) to abrade the surface. (Care must be taken to see that the surface is not touched by accident with the fingers because skin oils will be implanted onto the surface in this way).
- (iv) Use an acid etching agent on a gauze swab to clean the surface very thoroughly. Once again, care must be taken to handle the swab as little as possible to limit contamination with skin oils.
- (v) Always swab in one direction to avoid recontaminating the surface. Use the old swabs twice, at most (once each side). Then use a 4H pencil, if aluminium is to be marked, or a ball point pen if steel is to be marked, and draw lines on metal which will eventually coincide with the positioning marks on the strain gauges. The marks must be lightly drawn on the surface and not scribed into it. Next take the acid etching agent again and apply it to the surface by means of a cotton bud. Continue cleaning with cotton buds until there is no more discoloration. (You will need several buds).
- (vi) The area can now be considered clean enough for the strain gauge to be applied.
- (vii) Next, apply a neutraliser, normally a weak ammoniac solution, by means of a cotton bud and rub it vigorously over the surface. Then take a gauze swab and wipe the surface clean working in one direction only.

NB In the case of some high alloy steels there are problems with rapid oxidation of the surface directly after the neutralisation process. In such cases ignore steps (iv) & (v) above and proceed as follows:

- Apply acetone with a swab, first cleaning the area thoroughly. Once again take a clean swab and, working in one direction only, wipe the surface using very little acetone on the swab. (The wetter the surface is after it has been wiped, the greater

the likelihood that residue from the acetone will be left on it - even high-grade acetone for analytic purposes contains residue.)

- The area can now be considered clean enough for the strain gauge to be bonded.

6.2 Bonding the Strain Gauge:

- (i) Remove the strain gauge from the packet with the aid of tweezers (never touch the strain gauge with your fingers) and place it on a pre-cleaned sheet of glass. The strain gauge is placed on the glass just as it will be applied to the metal, i.e. surface to be bonded in contact with the glass.
- (ii) Affix the strain gauge to the glass with a length of sellotape. (In cases where the glue with which the strain gauge is to be applied is of the "baked-on" type, use Mylar adhesive tape in place of ordinary sellotape. This type of adhesive tape neither discolours nor burns at the temperature at which the glue is normally baked).
- (iii) Next starting from one point, pull the adhesive tape off the glass at a shallow angle. This prevents the strain gauge being bent through too sharp a radius and so ensures that it is not damaged.
- (iv) When the adhesive tape is free of the glass, take the tape in both hands and position the strain gauge carefully over the marks which have been made on the metal. Fix it securely with the aid of the tape already on it. (Step ii above).
- (v) Once again pull the sellotape off at a shallow angle, but now only until the strain gauge is free of the metal surface. Bend the free corner over and affix it lightly to the surface.

Should a "baked-on" glue be used the procedure will now be slightly different from that for a quickset glue. See instruction number (xi).

i.2.1 For quickset:

- (vi) Place a small drop of glue on the contact surface of the sellotape and the surface of the metal. If the quickset in question uses a catalyst to set, the catalyst should be smeared lightly over the surface of the strain gauge. NB: One wipe with nearly dry swab is enough. (Many of the quick-setting glues do not require a catalyst - the moisture in the air normally serves as a catalyst). Wait about one minute.
- (vii) Loosen the point of the sellotape that was originally lifted and place a polythene strip over the sellotape. (This prevents the fingers from being stuck to the metal by glue which has been forced out from under the sellotape.
- (viii) Now pass the index or middle finger firmly over the sellotape from the side where the glue has been placed. This ensures that there will eventually be no air bubbles trapped under the strain gauge. Stop when the finger is directly over the strain gauge.
- (ix) Maintain the pressure for one to two minutes, depending on the type of glue which is being used. Then remove the pressure.
- (x) Free one point the sellotape; bend it through 180° and remove it carefully from the strain gauge and the metal.

i.2.2 For Baked-on Glue:

- (xi) Spread the pre-prepared glue evenly over the surface where the strain gauge is to be applied.
- (xii) Fold the adhesive tape back over the surface in the original position. At this stage one can rub very lightly over the strain gauge with a small rubber to remove possible air bubbles.

- (xiii) Next place a soft rubber (e.g. Silicon rubber) pad on the strain gauge with a sturdy backing to support it. (The backing can e.g. be a plate or metal block). (The rubber should first be tested at maximum temperature.)
- (xiv) Now clamp the block firmly to the surface. (The clamping forces must be calculated from the pressure needed on the glue-consult the glue specifications.) Care must be taken that the strain gauge is not moved during the clamping process. (The Mylar-adhesive tape stretches rather a lot and it is quite possible for the strain gauge to be moved).
- (xv) Place it in an oven and heat it in accordance with the instructions that are provided with the glue.

7. THE K-FACTOR

The K-factor provides the relationship between the change in resistance, $\frac{\Delta R}{R}$, and strain when a strain gauge undergoes a specific strain. This may be expressed more simply by saying that the K-factor is an indication of the sensitivity of a strain gauge.

The K-factor indicated on a strain gauge packet is a weighted value of K_L and K_T , the K-factor in the lengthwise and transverse directions respectively.

For in the constantan strain gauges' in general use, K is nominally 2. The bridge amplifiers are then also designed as if $K = 2$.

The K values can however vary from approximately 1,95 to 2,25 for constantan strain gauges.

If a strain gauge should have K-factor greater than 2, it would mean that for a certain strain, ϵ , a greater change in resistance ratio, $\frac{\Delta R}{R}$, would be obtained. This would consequently result in a reading that is too high by a factor $\frac{K}{2}$. To compensate for this, the values may be multiplied by $\frac{2}{K}$ or the bridge amplifier's amplification can initially be adjusted to incorporate the value $\frac{2}{K}$.

When $K > 2$ the strain gauge is considered to be "too sensitive" and consequently the amplification of the bridge amplifier must be set lower. The same argument holds in the case of $K < 2$. In this case the strain gauge is "too insensitive" and must be compensated by increasing the amplification of the bridge amplifier.

8. SELF-HEATING EFFECT OF STRAIN GAUGES:

A strain gauge can be regarded as a small heating element. The most important factor in this case is firstly the resistance of the strain gauge and the grid area.

Take for example a strain gauge that has a nominal resistance of 120 ohms. Assume in addition that a constant voltage of 4 volts is applied across the two leads. The energy dissipation in the strain gauge is then

$$P = I^2 R = \frac{V^2}{R}$$
$$P = \frac{4^2}{120} = 0.8 \text{ W}$$
(6)

In a specific case with parameters as above, the dissipation is thus fixed value. (The influence of the change in resistance as a result of strain on the strain gauge is minimal): The heat that is generated in this way in the strain gauge is firstly conducted through the material to which it has been applied, and secondly to a lesser extent it is radiated from the strain gauge's surface.

The problems that arise as a result of this fact are:

- (i) If the strain gauge is bonded to a poor conductor, the strain gauge itself becomes heated which results in an apparent strain reading.
- (ii) In some materials the increase in temperature can result in a decrease in the modulus of elasticity which again would contribute to incorrect readings.

The steps that can be taken to combat this situation follow directly from the equation given above. We want to minimize the energy dissipation, P. This can be accomplished in two ways:

- (i) The voltage that is applied to strain gauge can be lowered.
- (ii) Alternatively the resistance of the strain gauge can be increased. (This can of course only be done during the original selection of the strain gauges for this particular application.
- (iii) A third method that may be used must be borne in mind when selecting the strain gauges.
- (iv)

This is to choose a strain gauge that has the greatest possible grid area. Conduction can take place more easily from a large surface than from a smaller surface. In this way therefore we can limit the energy density.

9. STRAIN GAUGE BRIDGE AMPLIFIERS:

Basically there are two types of bridge amplifiers, namely the D C amplifiers and the carrier wave amplifiers (A C amplifiers).

9.1 D C Amplifiers

In its simplest form the D C Amplifier appears as is shown in figure 8.

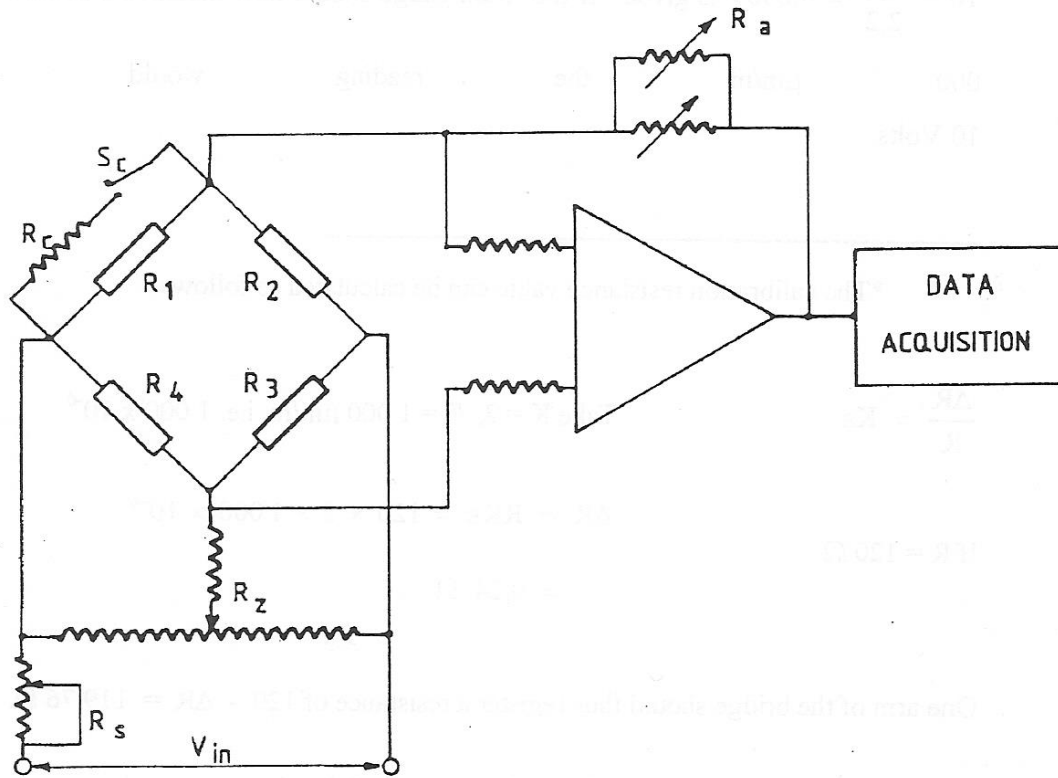


Figure 8. DC Amplifier circuit diagram

The use of this type of amplifier will be described very briefly, but how it works falls outside the scope of this course.

Assume that the strain gauge bridge consists of strain gauges R_1 , R_2 , R_3 and R_4 , in other words, it is in full bridge configuration.

After it has been switched on and a bridge voltage, V_{in} , has been decided upon, the unit's zero adjustment is made by means of a variable resistance, R_z . At this stage the amplification is set to minimum by means of R_a . (R_z can be made up of a coarse as well as a fine adjustment). The measuring apparatus must now have a zero reading.

Decide on a suitable coarse amplification. Set the bridge output at zero again.

Connect the calibration resistance*, R_c , with the switch S_c .

The amplifier now records an unbalance in the bridge and gives an output signal. If the K-factor of the strain gauge is not 2, the amplification of the bridge must be altered to compensate for this. (Suppose that the calibration resistance is equivalent to 1000 $\mu\text{m/m}$ and that this should represent a 10 Volt output. Suppose that $K = 2,2$. The amplification should

therefore be set so that an output reading of $10 \times 2.2/2 = 9,0909 \text{ V}$ is given. If the strain gauge should now measure a strain of $1000 \text{ } \mu\text{m/m}$, the reading would be 10 Volts.

*The calibration resistance value can be calculated as follows:

$$\frac{\Delta R}{R} = K\varepsilon$$

If R_1 is a $120 \text{ } \Omega$ strain gauge with gauge factor, $K = 2$, a calibration strain of $\varepsilon = 1000 \text{ } \mu\text{m/m}$, results in a change in resistance of:

$$\Delta R = RK\varepsilon = 120 \cdot 2 \cdot 1000 \times 10^{-6} = 0.24 \text{ } \Omega$$

One arm of the bridge should thus register a resistance of:

$$120 - \Delta R = 119,76 \text{ } \Omega.$$

The required resistance of the calibration resistor, R_c , can thus be calculated from the well-known parallel resistance equation

$$R_{equivalent} = \frac{R_c \cdot R_1}{R_c + R_1} = 119.76$$

From this it follows that

$$R_c = \frac{R_{eq} \cdot R_1}{R_1 - R_{eq}} = \frac{119.76 \cdot 120}{120 - 119.76} = 59880 \text{ } \Omega.$$

So, coupling a $59\,880 \text{ } \Omega$ resistor in parallel over the first resistor should give you an output reading equivalent to $1000 \text{ } \mu\text{m/m}$. This is shunt calibration.

An advantage of a DC amplifier is that very high frequency measurements can be made with it. A disadvantage involved in using this type of amplifier is the so-called thermocouple effect which can come into play at the soldered joints in the strain gauge circuit. (Because the current flows in only one direction, thermocouple voltage develops at the joints of dissimilar metals which influence the normal current in the conductors).

9.2 Carrier Wave Bridge Amplifiers:

This type of bridge is shown diagrammatically in Figure 9. Its operation can be briefly summarised as follows:

- (i) The signal generation provides a high frequency voltage which is supplied to the bridge.
- (ii) The bridge voltage adjustment is made just after this. This is followed by the bridge zero-adjustment. In this case it is a double action,
 - The zero-adjustment by means of resistances like those found in the D C bridge; and
 - the zero-adjustment by means of adjustable capacitor (to offset the effect of the reactance that arises as a result of the high frequency signal through the relatively long conductors, and also the capacitance between the conductors).
- (iii) Subsequently, the coarse as well as the fine adjustments can be made on the amplifier.

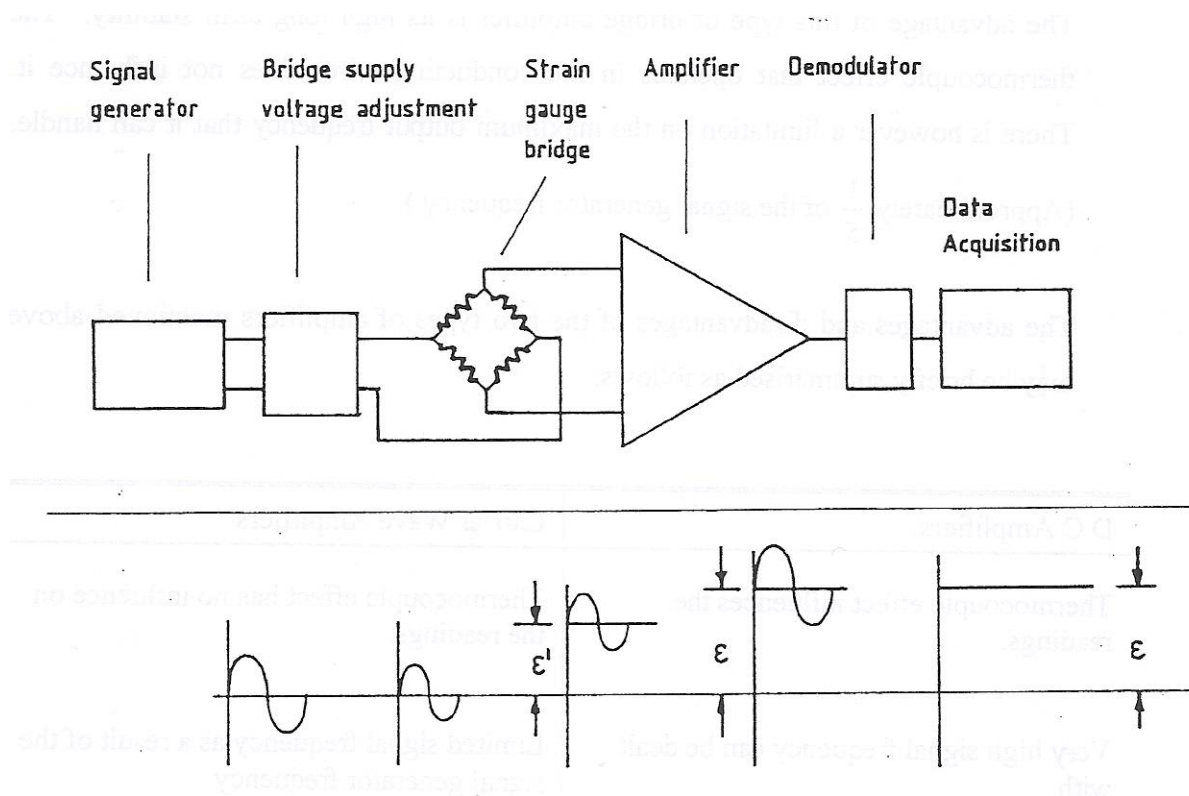


Figure 9. Carrier Wave Bridge Amplifier circuit diagram

At this stage the original modulated signal is still present and is thus eliminated by the so-called demodulator. The output from the demodulator is thus the functional signal that measuring apparatus etc. can be coupled to.

The advantage of this type of bridge amplifier is its high long term stability. The thermocouple effect that operates in the conducting wires does not influence it. There is however a limitation on the maximum output frequency that it can handle. The limiting frequency is approximately 1/5 of the signal generator frequency.

The advantages and disadvantages of the two types of amplifiers mentioned above may be briefly summarised as follows:

D C Amplifiers:	Carrier Wave Amplifiers.
Thermocouple effect influences the readings.	Thermocouple effect has no influence the readings.
Very high signal frequency can be dealt with.	Limited signal frequency as a result of the signal generator frequency.
Simple zero adjustment.	Zero adjustment more complex.
Long term stability not as high	High long term stability..
Low cost (?)	High cost (?)

10. TWO WIRE VERSUS THREE - WIRE WIRING.

If the bridge in Figure 10 is initially balanced no voltage will be measured between point C and D.

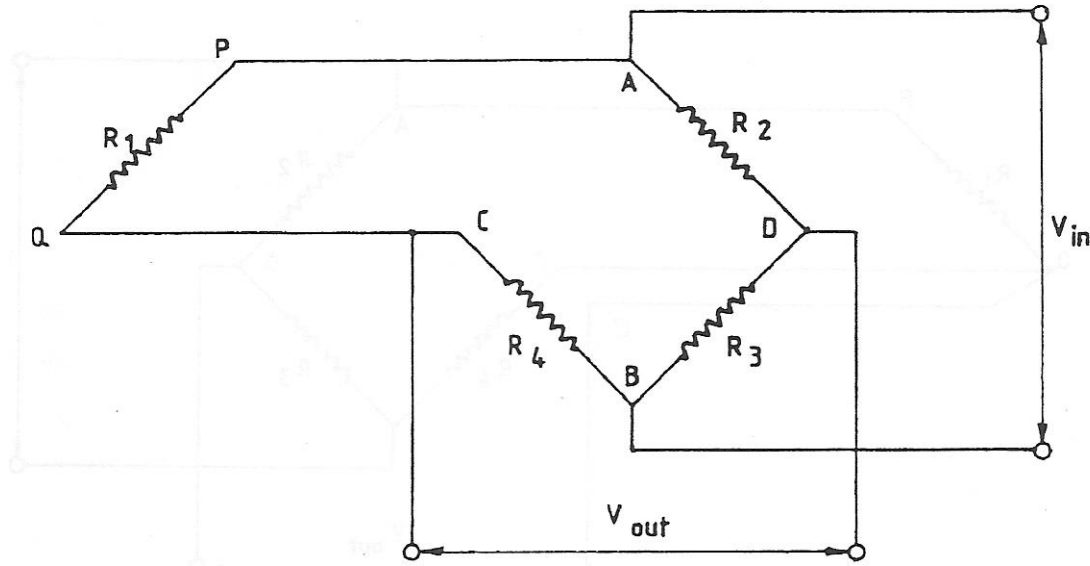


Figure 10

Suppose now that the resistance of all strain gauges increase by ΔR_n ($n = 1 \rightarrow 4$). The following equation will then apply for the increase in voltage between C and D.

$$\Delta V_{out} = V_{in} \frac{R_1 R_2}{(R_1 + R_2)^2} \left[\frac{\Delta R_1}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4} \right]$$

If the resistance of the conductors between AP and QC increase by the value ΔR_T , then

$$\Delta V_{out} = V_{in} \frac{R_1 R_2}{(R_1 + R_2)^2} \left[\frac{\Delta R_1 + 2\Delta R_T}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4} \right]$$

Supposing only R_1 is an active strain gauge and that $R_1 = R_2 = R_3 = R_4 = R$. The equation is thus simplified to:

$$\Delta V_{out} = V_{in} \frac{R_1 R_2}{(R_1 + R_2)^2} \left[\frac{\Delta R_1 + 2\Delta R_T}{R_1} \right] = V_{in} \frac{1}{4} \left[\frac{\Delta R + 2\Delta R_T}{R} \right]$$

There is thus a clearly discernible temperature-dependent effect on the output voltage.

The connection presented in Figure 11 can be investigated as an alternative.

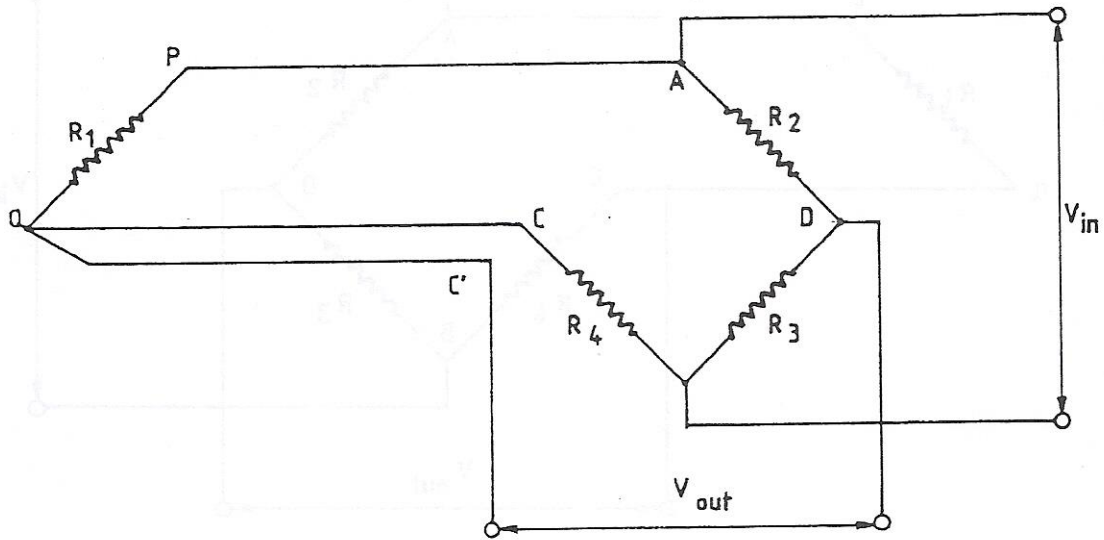


Figure 11

If the same reasoning is pursued for an increase in temperature in the conductors AP, QC and QC which consequently have an increase in resistance ΔR_T as a result, the equation 7 may be written as follows:

$$\Delta V_{out} = V_{in} \frac{R_1 R_2}{(R_1 + R_2)^2} \left[\frac{\Delta R_1 + 2\Delta R_T}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4 + 2\Delta R_T}{R_4} \right]$$

If $R_1 = R_2 = R_3 = R_4 = R$ again, and furthermore only R_1 is an active strain gauge, the above equation is simplified to:

$$\begin{aligned} \Delta V_{out} &= V_{in} \frac{R_1 R_2}{(R_1 + R_2)^2} \left[\frac{\Delta R_1 + 2\Delta R_T}{R_1} - \frac{2\Delta R_T}{R_4} \right] \\ &= V_{in} \frac{1}{4} \left[\frac{\Delta R + 2\Delta R_T}{R} - \frac{2\Delta R_T}{R} \right] \\ &= V_{in} \frac{1}{4} \left[\frac{\Delta R}{R} \right] \end{aligned}$$

From this it is very obvious that the effect of the long conductors undergoing changes in temperature can be eliminated by using a three-wire wiring system as in the case dealt with above.

One must bear in mind that R_1 must be kept at the same temperature as R_2, R_3 and R_4 . Furthermore the above equations naturally only hold true if $R_1 = R_2 = R_3 = R_4 = R$.

11. SIMPLE SCALE

If a load cell is manufactured, the output signal must be independent of the working line of the force that is applied. See the cantilever below.

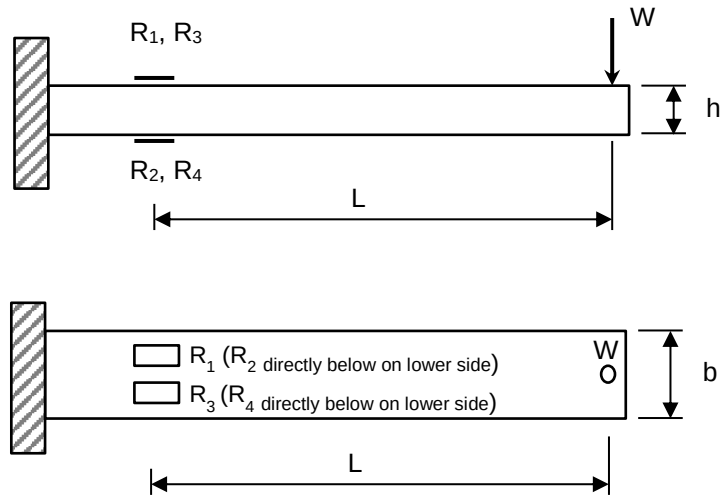


Figure 12

If a force W is applied, it will result in a strain at a distance L from the force, on the upper surface of:

$$\varepsilon_1, \varepsilon_3 = +\frac{6WL}{Ebh^2}$$

(8)

and at the same position on the lower surface it will be:

$$\varepsilon_2, \varepsilon_4 = -\frac{6WL}{Ebh^2} .$$

(9)

It is clear from this that the readings obtained from R_1, R_3 , and R_2, R_4 are dependent on L , the distance between the line of force and the strain gauges.

This situation can, however, be improved through the following connection:

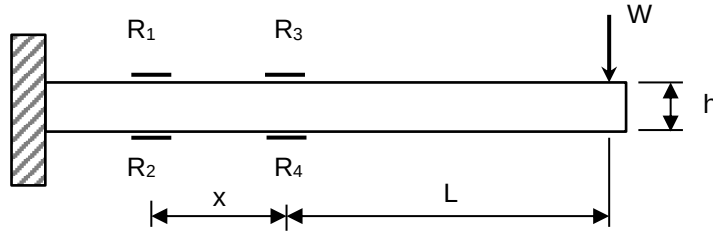


Figure 13

Consider only the positive strains i.e. ε_1 and ε_3

$$\varepsilon_1 = + \frac{6W(L+x)}{Ebh^2}$$

$$\varepsilon_3 = + \frac{6WL}{Ebh^2}$$

The strain gauges can now be wired so that the difference ($\varepsilon_1 - \varepsilon_3$) is measured. This provides the following equation for the upper side of the cantilever.

$$\begin{aligned} \varepsilon_1 - \varepsilon_3 &= \frac{6W(L+x)}{Ebh^2} - \frac{6WL}{Ebh^2} \\ &= \frac{6W(L+x-L)}{Ebh^2} \\ &= \frac{6Wx}{Ebh^2} \end{aligned}$$

(10)

The output signal of the bridge is thus independent of the distance from the line of force to the strain gauges with one important proviso namely that the force must not be applied between the two strain gauges.

This type of scale is relatively insensitive and in practice other configurations are used.

12. ROSETTE ANALYSIS.

So-called rosette strain gauges are often used in experimental stress analysis. See figure 14.

This so-called rosette usually consists of at least three strain gauges arranged in a certain pattern. Using these three strain gauges it is possible to calculate the two principal stresses from the strains obtained from the strain gauges, as long as the Modulus of Elasticity or Young's Modulus and the Poisson's coefficient of the metal to which the rosette is bonded is known. Furthermore the angle that the principal stress makes with one of the strain gauges can also be calculated.

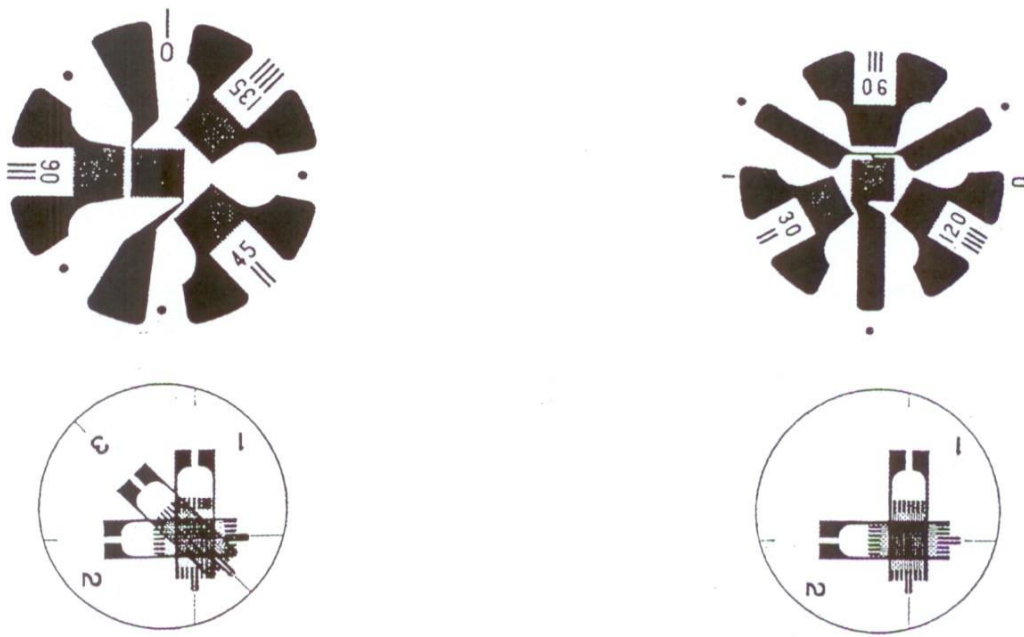
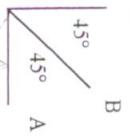
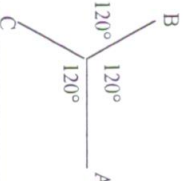
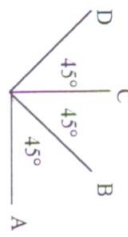
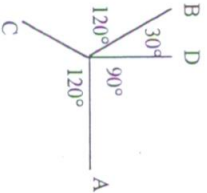


Figure 14. Various strain gauge rosettes

In the accompanying table the equations are given that are applicable to the various rosette configurations. The principal strains and the principal stresses as well as the angle that the maximum principal stress makes with one of the strain gauges can be calculated.

Rosette type	Principal strain and Principal Stresses	Principal Stress Angle	Identification $0 < \beta < 90^\circ$
Three-element rectangular $2 \times 90^\circ + 45^\circ$ 	$\epsilon_{1,2} = \frac{\epsilon_A + \epsilon_C}{2} \pm \frac{1}{2} \sqrt{(\epsilon_A - \epsilon_C)^2 + (2\epsilon_B - \epsilon_A - \epsilon_C)^2}$ $\sigma_{1,2} = \frac{E}{2} \left[\frac{\epsilon_A + \epsilon_C}{1 - \nu} \pm \frac{1}{1 + \nu} \sqrt{(\epsilon_A - \epsilon_C)^2 + (2\epsilon_B - \epsilon_A - \epsilon_C)^2} \right]$	$\tan 2\beta = \frac{2\epsilon_B - \epsilon_A - \epsilon_C}{\epsilon_A - \epsilon_C}$	$\epsilon_B > \frac{\epsilon_A + \epsilon_C}{2}$
Delta $3 \times 120^\circ$ 	$\epsilon_{1,2} = \frac{\epsilon_A + \epsilon_B + \epsilon_C}{3} \pm \frac{\sqrt{2}}{3} \sqrt{(\epsilon_A - \epsilon_B)^2 + (\epsilon_B - \epsilon_C)^2 + (\epsilon_C - \epsilon_A)^2}$ $\sigma_{1,2} = \frac{E}{3} \left[\frac{\epsilon_A + \epsilon_B + \epsilon_C}{(1 - \nu)} \pm \frac{\sqrt{2}}{(1 + \nu)} \sqrt{(\epsilon_A - \epsilon_B)^2 + (\epsilon_B - \epsilon_C)^2 + (\epsilon_C - \epsilon_A)^2} \right]$	$\tan 2\beta = \frac{\sqrt{3}(\epsilon_C - \epsilon_B)}{2\epsilon_A - (\epsilon_B + \epsilon_C)}$	$\epsilon_C > \epsilon_B$
Four-element rectangular (Very seldom used) 	$\epsilon_{1,2} = \frac{\epsilon_A + \epsilon_B + \epsilon_C + \epsilon_D}{4} \pm \frac{1}{2} \sqrt{(\epsilon_A - \epsilon_C)^2 + (\epsilon_B - \epsilon_D)^2}$ $\sigma_{1,2} = \frac{E}{2} \left[\frac{\epsilon_A + \epsilon_B + \epsilon_C + \epsilon_D}{2(1 - \nu)} \pm \frac{1}{1 + \nu} \sqrt{(\epsilon_A - \epsilon_C)^2 + (\epsilon_B - \epsilon_D)^2} \right]$	$\tan 2\beta = \frac{(\epsilon_B - \epsilon_D)}{(\epsilon_A - \epsilon_C)}$	$\epsilon_B > \epsilon_D$
T-Delta 	$\epsilon_{1,2} = \frac{\epsilon_A + \epsilon_D}{2} \pm \frac{1}{2} \sqrt{(\epsilon_A - \epsilon_D)^2 + \frac{4}{3}(\epsilon_C - \epsilon_B)^2}$ $\sigma_{1,2} = \frac{E}{2} \left[\frac{\epsilon_A + \epsilon_D}{2(1 - \nu)} \pm \frac{1}{1 + \nu} \sqrt{(\epsilon_A - \epsilon_D)^2 + \frac{4}{3}(\epsilon_C - \epsilon_B)^2} \right]$	$\tan 2\beta = \frac{2(\epsilon_C - \epsilon_B)}{\sqrt{3}(\epsilon_A - \epsilon_D)}$	$\epsilon_C > \epsilon_D$

13. RESIDUAL STRESS ANALYSIS.

It is often necessary to determine residual stresses in parts or structures. Various techniques have been developed over the years, but they are destructive (meaning you have to cut into the structure, destroying it, in order to evaluate the residual stress).

Using the information that has already been covered in connection with rosette strain gauge analysis, a method has been developed that can be described as semi-destructive.

In this method, a very small hole, 1,5 mm to 5 mm, is milled in the part in question to a depth approximately equal to the diameter of the hole. As a result the stress in the immediate vicinity of the hole is relieved. This change in the stress pattern in the immediate vicinity of the hole can be measured with the aid of a rosette strain gauge.

Special strain gauges have been developed for this purpose. Normally the three different gauges of the rosette are arranged around a central point. See figure 15.

The method of analysis is as follows; an end-mill is positioned centrally relative to the three strain gauges of the rosette. The hole is then milled in increments. The strain readings of the three strain gauges are recorded at every depth. Principal stresses can be calculated for each set of strain values obtained. At some depth, maxima will be reached which are taken as the final values.

It can be stated with reference to figure 15 that if strain gauge A has a reading of ε_A and B a reading of ε_B etc., the two principal stresses can be determined by:

$$\sigma_1 = \frac{S}{4A} + \frac{D}{4b \cos 2\beta} \quad (11)$$

$$\sigma_2 = \frac{S}{4A} - \frac{D}{4b \cos 2\beta} \quad (12)$$

$$\tan 2\beta = \frac{\varepsilon_A + \varepsilon_C - \varepsilon_B}{\varepsilon_A - \varepsilon_C} \quad (13)$$

$$S = \varepsilon_A + \varepsilon_C \quad (14)$$

$$D = \varepsilon_A - \varepsilon_C \quad (15)$$

$$A = \frac{-(1 + \nu)}{2Er^2} \quad (16)$$

$$B = \frac{3(1 + \nu) - 4r^2}{2Er^2} \quad (17)$$

$$r = \frac{R}{R_0}$$

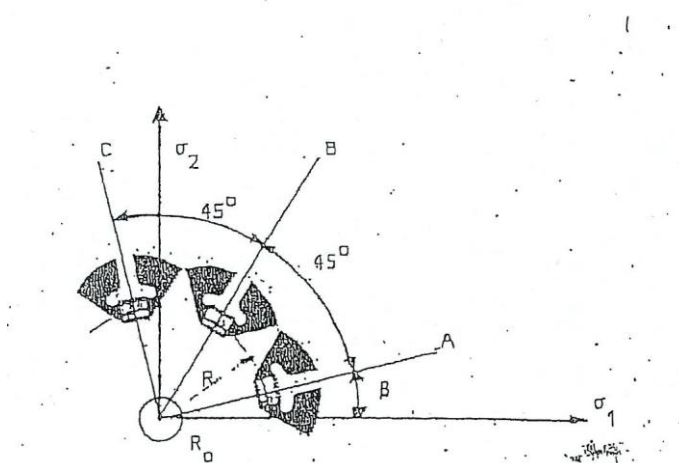


Figure 15

where R_0 is the radius of the hole as shown in Figure 15.

14. TORQUE MEASUREMENT WITH FOUR STRAIN GAUGES.

The strain gauges should be mounted at 45° to the longitudinal axis of the shaft as shown in Figure 16.

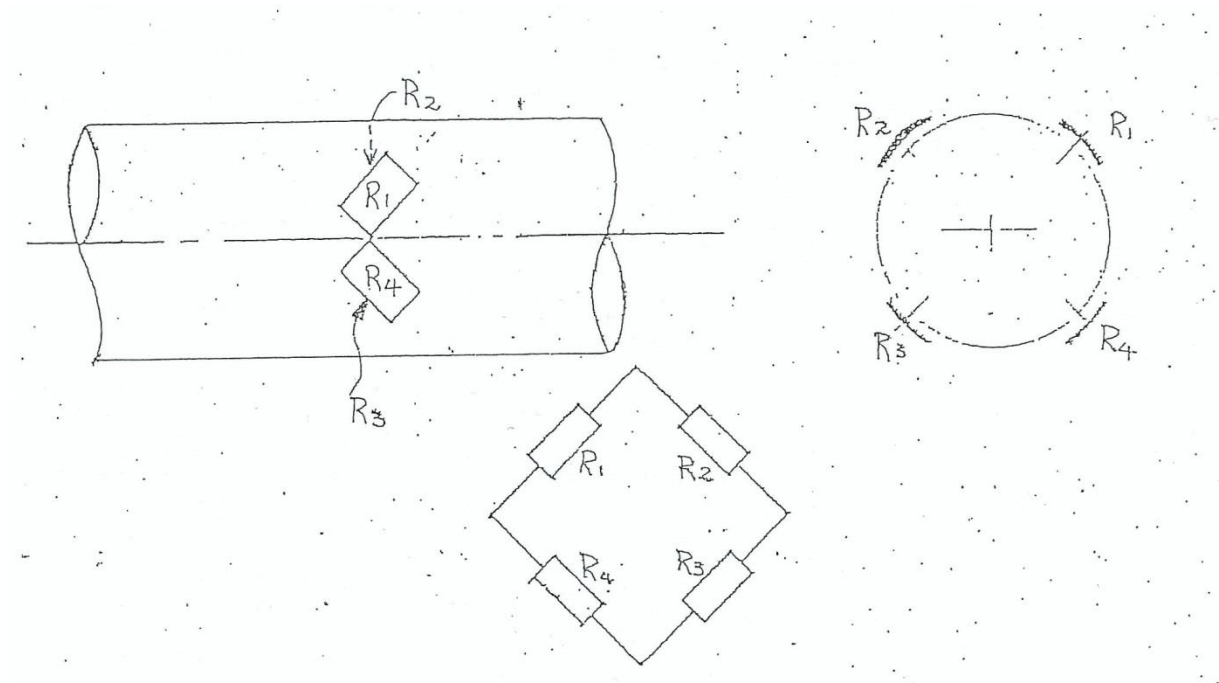


Figure 16

Each strain gauge will give a readout equal to a quarter of the total readout.

$$\varepsilon'_1 = \varepsilon'_2 = \varepsilon'_3 = \varepsilon'_4 = \varepsilon'$$

The total readout will then be:

$$\varepsilon'_1 + \varepsilon'_2 + \varepsilon'_3 + \varepsilon'_4 = 4\varepsilon' = \varepsilon$$

The torque experienced by the shaft can then be calculated from

$$T = \frac{\pi D^3 G}{32}$$

where

T = Torque in Newton meters

D = Shaft diameter in meters.

G = Shear modulus of shaft material (i.e. 82×10^9 Newton per metre for steel)

E = total strain readout

$$T = 7.91 \times 10^9 D^3 \varepsilon$$

EXAMPLE:

Say for instance a steel shaft of 90 mm diameter has been fitted with strain gauges. It was then coupled via slip rings to a bridge amplifier. The amplifier was calibrated to give 10 Volt output for 5 000 micro strain input signal. Under normal working conditions of the shaft, it was found that the output was 8,3 Volts. What is the torque in the shaft?

$$5\,000 \text{ microstrain} + 10 \text{ Volts}$$

$$8.3 \text{ Volts} + \frac{8.3}{10} \times 5\,000 \text{ microstrain} = 4150 \text{ microstrain}$$

(Although it is not necessary to know this in this particular example, each strain gauge contributes

$$\frac{4150}{4} \text{ microstrain to the total, that is } 1037.5 \text{ microstrain}$$

Mild steel, yields in the vicinity of about 1200 *microstrain*.

The 4150 *microstrain* is in fact

$$4150 \times 10^{-6} \text{ m/m}$$

Therefore,

$$T = 7.91 \times 10^9 (0.09)^3 4150 \times 10^{-6} = 23930 \text{ Nm} = 23.9 \text{ kNm}$$

REFERENCES :

- 1) Thomson, W (Lord Kelvin): On the Electrodynamic Qualities of Metals. Phil. Trans. Roy.Soc.(London). Vol 146, pp 649 - 751, 1856.
- 2) Perry,CC; Lissner,HR The Strain Gauge Primer 1stEd: pp 15-28.
- 3) Dally,JW; Riley, WF: Experimental Stress Analysis 2ndEd pp 154-155.

USEFUL BOOKS WHICH DEAL WITH STRAIN GAUGES AND THEIR USES:

- 1) (Two of the books mentioned are old and may be out of print, but the treatment or certain basic principles is very good). (They will usually be found in technical libraries).

Daily J W & Riley W F: Experimental Stress Analysis, (2nd Edition) McGraw Hill.
- 2) Perry CC & Lissner HR: The Strain Gauge Primer. McGraw Hill.
- 3) Potma T: Strain Gauges, Philips.